Automorphism Ratios of Finite Groups

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Abstract

For a finite group G, set $\alpha(G) = |\mathrm{Aut}(G)|/|G|$. We prove that every positive rational number occurs as $\alpha(G)$, both for finite groups in general and for finite nilpotent groups. The first proof is constructive: for $r = a/b$, a perfect group of ratio $1/(2b)$ is paired with a metacyclic group of coprime order and ratio $2a$. The nilpotent case follows from a local result. For each prime p, there is an integer m_p, coprime to p, such that every value $m_p p^k$, with $k \in \mathbb{Z}$, is realized by a finite p-group. Outside a fixed finite set, every prime divisor of m_p is smaller than p. This support condition allows the local factors to be assembled in a finite direct product. Finally, every fiber of α is countably infinite and contains groups of unbounded order and groups with arbitrarily many nonabelian composition factors.

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1 Introduction

Throughout, $\mathbb{Q}^+$ denotes the positive rationals and $\mathbb{Z}^+$ the positive integers. For a finite group G, let

$$\alpha(G) = \frac{|\mathrm{Aut}(G)|}{|G|}.$$

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For $x \in \mathbb{Q}^+$, let $\operatorname{supp}(x)$ be the set of primes q for which $v_q(x) \neq 0$. Problem 21.97 of the Kourovka Notebook, attributed to M. Tărnăuceanu, asks whether every element of $\mathbb{Q}^+$ occurs as $\alpha(G)$ for some finite group G [KM26].

There are several partial results. For finite abelian groups, Tărnăuceanu proved that the set of values $\alpha(A)$ is dense in $[0, \infty)$ [Tăr26]. It does not, however, contain every positive rational. McCulloch proved that if a/b is in lowest terms and $\alpha(A) = a/b$ for a finite abelian group A, then b is squarefree. He also proved that no odd prime occurs as such a value [McC26]. The structure of automorphism groups of finite abelian groups goes back to Ranum [Ran07]. A modern account is given by Hillar and Rhea [HR07]. In a different direction, González-Sánchez and Jaikin-Zapirain constructed, for every prime p, a sequence of nonabelian finite p-groups G_i for which $|\operatorname{Aut}(G_i)|/|G_i| \to 0$ [GJ15]. These results do not determine the image of α.

Two elementary examples indicate the main difficulties. For a cyclic group,

$$\alpha(C_n) = \frac{\varphi(n)}{n} = \prod_{p \mid n} \frac{p-1}{p},$$

so the exponents in n have no effect and every nontrivial cyclic value is less than 1. Direct products offer more flexibility, but α is not multiplicative in general: $\alpha(C_2 \times C_2) = 3/2$, whereas $\alpha(C_2)^2 = 1/4$. The discrepancy comes from automorphisms that mix isomorphic direct factors. Coprime orders rule out this mixing and will be sufficient for the finite construction. A second useful identity is

$$\alpha(G) = \frac{|\operatorname{Out}(G)|}{|Z(G)|}.$$

It turns denominator control into a question about centers and outer automorphisms. Since enlarging the center can also produce new central automorphisms, we use quotients of universal central extensions of perfect centerless groups.

We answer Problem 21.97 [KM26] and the nilpotent version asked by McCulloch [McC26].

Theorem 1.1. *(a) For every $r \in \mathbb{Q}^+$, there is a finite group G, constructed from r, such that $\alpha(G) = r$.*

(b) For every $r \in \mathbb{Q}^+$, there is a finite nilpotent group N such that $\alpha(N) = r$. It follows that the images of α on finite supersolvable groups and finite solvable groups are also $\mathbb{Q}^+$.

(c) For every $r \in \mathbb{Q}^+$, the set of isomorphism classes of finite groups G with $\alpha(G) = r$ is countably infinite. They may be chosen to have unbounded order and arbitrarily many nonabelian composition factors.

Although part (a) follows from part (b), the separate finite construction is useful: it starts directly from the prime factorization of r and avoids the local p-group argument. Its denominator factors come from perfect central quotients. For odd p, a Jacobi group has ratio $(p-1)/p$. The primes introduced through $p-1$ are smaller than p and can be removed recursively. A separate construction supplies every ratio 2^{-e} with $e \geq 0$. Together these give, for each positive integer b, a finite perfect group D_b with $\alpha(D_b) = 1/b$. A metacyclic group supplies the required even integer ratio $2a$ while avoiding the primes dividing the denominator factor. Applied with ratios $1/(2b)$ and $2a$, the two groups have coprime orders and their direct product has ratio a/b.

The nilpotent case requires a different construction. For a prime p, let

$$\mathcal{I}_p = \{\alpha(P) \mid P \text{ is a finite } p\text{-group}\}.$$

The local input is the following.

Theorem 1.2. *For every prime p, there is a positive integer m_p, coprime to p, such that $m_p p^{\mathbb{Z}} \subseteq \mathcal{I}_p$. The integers m_p may be chosen so that there is a finite set of primes $\mathcal{E}$, independent of p, with*

$$\operatorname{supp}(m_p) \subseteq \mathcal{E} \cup \{q \mid q < p\}.$$

The support condition is what makes the local groups assemble. Starting with the primes occurring in a prescribed rational number, close the set under $p \mapsto \operatorname{supp}(m_p)$. Every newly introduced prime is either in the fixed set $\mathcal{E}$ or is smaller than the prime that introduced it, so this closure is finite. The exponents in the lines $m_p p^{\mathbb{Z}}$ can then be chosen to match every valuation of the prescribed rational.

Section 2 gives the direct finite construction. In Section 3, the local lines are built from finite quotients of the González-Sánchez and Jaikin-Zapirain pro-p groups together with auxiliary finite p-groups, and their supports are controlled through the associated rational Lie algebra. The local-to-global argument then proves part (b). Finally, Section 4 proves part (c) by adjoining suitable complete simple direct factors without changing α.

2 A finite construction

2.1 Central extensions and automorphism ratios

All groups in this section are finite. We write $Z(G)$, $\operatorname{Inn}(G)$, and $\operatorname{Out}(G) = \operatorname{Aut}(G)/\operatorname{Inn}(G)$ for the center, inner automorphism group, and outer automorphism group of G, respectively.

Lemma 2.1. *For every finite group G, one has $\alpha(G) = |\operatorname{Out}(G)|/|Z(G)|$.*

Proof. Since $|\operatorname{Inn}(G)| = |G|/|Z(G)|$ and $|\operatorname{Aut}(G)| = |\operatorname{Inn}(G)||\operatorname{Out}(G)|$, the formula follows. □

We use the homological convention $M(Q) = H_2(Q, \mathbb{Z})$ for the Schur multiplier. If Q is perfect, the kernel of its universal central extension is naturally identified with $M(Q)$. See [Bro82] and [Kar87].

Lemma 2.2. *Let Q be a finite perfect group, and let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be its universal central extension. Then E is perfect and $M \le Z(E)$. If Q is centerless, then $Z(E) = M$. Every $\sigma \in \operatorname{Aut}(Q)$ has a unique lift $\widetilde{\sigma} \in \operatorname{Aut}(E)$ satisfying $\pi\widetilde{\sigma} = \sigma\pi$. The induced action of $\operatorname{Aut}(Q)$ on M factors through $\operatorname{Out}(Q)$.*

Proof. The extension is central, so $M \le Z(E)$. We first check that E is perfect. Since Q is perfect, $\pi(E') = Q$. Let i denote the inclusion of E' in E. The restriction $\pi|_{E'}$ is a central extension of Q. Universality gives a homomorphism ϕ from E to E' with $(\pi|_{E'})\phi = \pi$. The composite $i\phi$ is an endomorphism of E over Q, so uniqueness gives $i\phi = \mathrm{id}_E$. This makes i surjective and proves that $E = E'$.

If Q is centerless and $x \in Z(E)$, then $\pi(x) \in Z(Q)$, so $x \in M$. Thus $Z(E) = M$.

For $\sigma \in \mathrm{Aut}(Q)$, let $\pi_\sigma = \sigma^{-1} \circ \pi$, viewed as a map from E to Q. This is a central extension. Universality gives a unique endomorphism $\widetilde{\sigma}$ of E with $\pi_\sigma \widetilde{\sigma} = \pi$, equivalently $\pi \widetilde{\sigma} = \sigma \pi$. The unique lift of σ^{-1} is inverse to $\widetilde{\sigma}$, by uniqueness for $\sigma = \mathrm{id}_Q$. Hence $\widetilde{\sigma}$ is an automorphism. The same uniqueness gives $\widetilde{\sigma\tau} = \widetilde{\sigma}\widetilde{\tau}$, and each $\widetilde{\sigma}$ preserves M.

Suppose that σ is conjugation by $q \in Q$, and choose $e \in E$ with $\pi(e) = q$. Conjugation by e is the unique lift of σ. It fixes M pointwise because $M \leq Z(E)$. Inner automorphisms of Q therefore act trivially on M, and the action factors through $\mathrm{Out}(Q)$. □

Lemma 2.3. *Let Q be finite, perfect, and centerless. Let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be its universal central extension, and let $N \leq M$. If $G = E/N$, then $Z(G) = M/N$ and $\mathrm{Out}(G) \cong \mathrm{Stab}_{\mathrm{Out}(Q)}(N)$, where the stabilizer is taken for the action on M from Theorem 2.2. Consequently*

$$\alpha(G) = \frac{|\mathrm{Stab}_{\mathrm{Out}(Q)}(N)|}{|M/N|}.$$

Proof. Let ρ be the quotient map from E to G, and let $\bar{\pi}$ be the induced map from G to Q. The inclusion $M/N \leq Z(G)$ is immediate. If $xN \in Z(G)$, then $[x, E] \leq N$, so $\pi(x) \in Z(Q) = 1$. Hence $x \in M$, and $Z(G) = M/N$.

By Theorem 2.2, the group E is perfect, so G is perfect. Every automorphism of G preserves $Z(G)$ and therefore induces an automorphism of Q through $\bar{\pi}$. If $\theta \in \mathrm{Aut}(G)$ induces the identity on Q, then $\theta(g) = gz_g$ with $z_g \in Z(G)$ for every $g \in G$. Thus θ fixes all commutators, and $G = G'$ forces $\theta = 1$. The map $\mathrm{Aut}(G) \to \mathrm{Aut}(Q)$ is therefore injective.

Let $A \leq \mathrm{Aut}(Q)$ be the subgroup of automorphisms whose unique lifts to E stabilize N. If $\sigma \in A$, then $\widetilde{\sigma}$ descends to an automorphism of G. Thus A lies in the image.

Conversely, let $\varphi \in \mathrm{Aut}(G)$ induce $\beta \in \mathrm{Aut}(Q)$. Both $\varphi\rho$ and $\rho\widetilde{\beta}$ are morphisms from E to the central extension $(G, \beta^{-1}\bar{\pi})$ over Q. Indeed, $(\beta^{-1}\bar{\pi})\varphi\rho = \pi = (\beta^{-1}\bar{\pi})\rho\widetilde{\beta}$. The universal property therefore gives $\varphi\rho = \rho\widetilde{\beta}$. For $n \in N$, this gives $1 = \varphi\rho(n) = \rho\widetilde{\beta}(n)$, so $\widetilde{\beta}(N) \leq N$. Since N is finite, $\widetilde{\beta}(N) = N$. Hence the image is A, and $\mathrm{Aut}(G) \cong A$.

The subgroup $\mathrm{Inn}(G)$ maps isomorphically onto $\mathrm{Inn}(Q)$, because $G/Z(G) \cong Q$ and Q is centerless. Inner automorphisms of Q lie in A, since their lifts are conjugations in E and $N \leq Z(E)$. The group A is the preimage of $\mathrm{Stab}_{\mathrm{Out}(Q)}(N)$ under $\mathrm{Aut}(Q) \to \mathrm{Out}(Q)$, because inner automorphisms act trivially on M. Therefore $\mathrm{Out}(G) \cong \mathrm{Stab}_{\mathrm{Out}(Q)}(N)$. The formula for $\alpha(G)$ follows from Theorem 2.1 and the center computation above. □

Corollary 2.4. *If Q is finite, perfect, and centerless, and $E \to Q$ is its universal central extension with kernel M, then $\alpha(E) = |\mathrm{Out}(Q)|/|M|$.*

Proof. Apply Theorem 2.3 with $N = 1$. □

Lemma 2.5. *Let $Q_1, \ldots, Q_t$ be perfect groups, and let $\widehat{Q}_i \to Q_i$ be their universal central extensions. Then $\prod_i \widehat{Q}_i \to \prod_i Q_i$ is the universal central extension of $\prod_i Q_i$.*

Proof. By induction, it is enough to treat two factors. The product map is a central extension, and its source is perfect. Let η be a central extension from X to $Q_1 \times Q_2$. Let $X_i = \eta^{-1}(Q_i)$, where Q_i is identified with the corresponding direct factor of $Q_1 \times Q_2$. Then the map from X_i to Q_i is a central extension. Universality gives unique maps from $\widehat{Q}_i$ to X_i over the two factors. Their mutual commutators lie in $\ker \eta$. For a fixed element of either image, commutation with it is a homomorphism from the other $\widehat{Q}_i$ to the abelian group $\ker \eta$. Perfectness forces this homomorphism to be trivial, so the images commute and the two maps combine to a homomorphism from $\widehat{Q}_1 \times \widehat{Q}_2$ to X over $Q_1 \times Q_2$. Uniqueness follows by restriction to the two factors. □

Lemma 2.6. *Let $G = G_1 \times \cdots \times G_t$, where each G_i is perfect and $G_i/Z(G_i)$ is centerless. Set $Q_i = G_i/Z(G_i)$. If every automorphism of $Q_1 \times \cdots \times Q_t$ preserves each direct factor Q_i, then every automorphism of G preserves each G_i. Hence $\mathrm{Aut}(G) \cong \prod_i \mathrm{Aut}(G_i)$ and $\alpha(G) = \prod_i \alpha(G_i)$.*

Proof. The center of G is $\prod_i Z(G_i)$, so $G/Z(G) \cong \prod_i Q_i$. Let P_i be the preimage of the ith direct factor Q_i in G. Then $P_i = G_i \times \prod_{j \neq i} Z(G_j)$. Any automorphism of G induces an automorphism of $G/Z(G)$, hence preserves each P_i. Since $P_i' = G_i$, it preserves each G_i. Restriction gives the product decomposition of automorphism groups, and the formula for α follows. □

Lemma 2.7. *If A and B are finite groups with $\gcd(|A|, |B|) = 1$, then $\mathrm{Aut}(A \times B) \cong \mathrm{Aut}(A) \times \mathrm{Aut}(B)$ and $\alpha(A \times B) = \alpha(A)\alpha(B)$.*

Proof. The factor $A \times 1$ is precisely the set of elements whose orders have all prime divisors among the primes dividing $|A|$. This set is characteristic. The same argument gives that $1 \times B$ is characteristic. Restriction to the two factors gives the result. □

2.2 Graph codes and stabilizers

If isomorphic factors are repeated, automorphisms may permute the corresponding multiplier coordinates. Thus repeated factors of ratio $1/2$ do not by themselves produce arbitrary powers of $1/2$. The graph code below is used to choose a quotient subspace with trivial stabilizer under the relevant coordinate action.

Let $\Gamma = (V, E)$ be a finite simple graph. Choose an orientation of every edge. For a field F, define a linear map F_Γ from F^V to $\bigoplus_{e \in E} F^2$ by sending $x = (x_v)$ to the vector whose coordinate at an oriented edge from u to v is (x_u, x_v). Let $U_\Gamma = \mathrm{im}(F_\Gamma)$. Changing orientations conjugates the construction by coordinate swaps inside the edge summands.

Lemma 2.8. *Suppose that Γ has minimum degree at least 2. Then*

$$\mathrm{Stab}_{\mathrm{GL}_2(F)^E \rtimes S_E}(U_\Gamma) \cong (F^\times)^V \rtimes \mathrm{Aut}(\Gamma),$$

where $\mathrm{Aut}(\Gamma)$ acts on $(F^\times)^V$ by permuting vertices.

Proof. Since Γ has no isolated vertices, F_Γ is injective. Let g stabilize U_Γ, and define $T \in \mathrm{GL}(F^V)$ by $gF_\Gamma = F_\Gamma T$. Let ϵ_v be the coordinate functional attached to v. For an edge $e = \{u, v\}$, set $L_e = \langle \epsilon_u, \epsilon_v \rangle$. Since g^* permutes the dual edge summands and $F_\Gamma^* g^* = T^* F_\Gamma^*$, the map T^* permutes the planes L_e.

For distinct edges e and f, the intersection $L_e \cap L_f$ is nonzero exactly when the edges share a vertex. In that case the intersection is the coordinate line of the shared vertex. Here the minimum-degree hypothesis enters. Every coordinate line occurs as one of these intersections, so their permutation by T^* has the form $T^*(\epsilon_v) = \lambda_v \epsilon_{\tau(v)}$ for scalars $\lambda_v \in F^\times$ and a permutation τ of V. Preservation of the planes L_e makes τ an automorphism of Γ.

The pair (λ, τ) determines T. It also determines the edge permutation of g, since the graph is simple. Finally, surjectivity of each projection $U_\Gamma \to F^2$ determines the corresponding edge matrix. Conversely, vertex scalars and graph automorphisms act edgewise, with a coordinate swap when the chosen orientations disagree, and stabilize U_Γ. This identifies the stabilizer with the stated semidirect product. □

Corollary 2.9. *If Γ is asymmetric and has minimum degree at least 2, then the stabilizer of U_Γ in $\mathrm{GL}_2(2)^E \rtimes S_E$ is trivial.*

Proof. In $\mathbb{F}_2$ the scalar group $(\mathbb{F}_2^\times)^V$ is trivial. Apply Theorem 2.8. □

Construction 2.10. For $e \geq 3$, let Γ_e be obtained by stringing together $\Theta(1,2,3)$, $e-3$ triangles, and $\Theta(1,2,4)$, in that order. Here $\Theta(a,b,c)$ consists of two branch vertices joined by internally disjoint paths of lengths a, b, and c. Order the branch vertices of each theta graph as left and right terminals, choose two adjacent terminals in every triangle, and identify the right terminal of each piece with the left terminal of the next.

When $e = 3$, there are no triangles between the two theta pieces, which meet at one terminal. Thus Γ_3 has 13 edges and 10 vertices, and $|V(\Gamma_3)| - |E(\Gamma_3)| = -3$, as required for the first exponent in Theorem 2.14.

Lemma 2.11. *For $e \geq 3$, the graph Γ_e is asymmetric, has minimum degree at least 2, and satisfies $|E(\Gamma_e)| = 3e + 4$ and $|V(\Gamma_e)| = 2e + 4$.*

Proof. The pieces are precisely the maximal 2-connected subgraphs of Γ_e, and the successive identifications are precisely its cut vertices. Every automorphism therefore acts on the path of pieces. The nonisomorphic endpoints rule out reversal of this path, leaving every piece and cut vertex fixed. In an endpoint theta piece, the fixed cut vertex determines both branch vertices, and the three distinct path lengths fix the piece pointwise. The two terminal vertices likewise fix each intervening triangle pointwise. Every vertex of Γ_e is fixed, proving that $\mathrm{Aut}(\Gamma_e) = 1$.

All vertices have degree at least 2 before gluing, and gluing only increases degrees. Finally $|E(\Gamma_e)| = 6 + 7 + 3(e-3) = 3e + 4$, and $|V(\Gamma_e)| = 5 + 6 + 3(e-3) - (e-2) = 2e + 4$. □

2.3 Numerator and denominator factors

We first construct groups with a prescribed integer ratio, subject to the parity condition below, while avoiding a specified finite set of primes in their order. The prime-avoidance clause will allow the numerator factor to be combined with the denominator factor by a coprime direct product.

Lemma 2.12. *Let $d \geq 1$, and let Π be a finite set of primes. Assume that $2 \in \Pi$ implies $2 \mid d$. Then there is a finite group A such that $\alpha(A) = d$ and no prime in Π divides $|A|$.*

Proof. Write $d = \prod_{r \mid d} r^{a_r}$. We choose a prime q by imposing one congruence for each relevant prime r. If $r \mid d$ and $r \notin \Pi$, require

$$q \equiv 1 \pmod{r^{a_r}}.$$

If $r \mid d$ and $r \in \Pi$, require

$$q \equiv 1 + d \pmod{r^{a_r+1}}.$$

Finally, if $r \in \Pi$ and $r \nmid d$, then r is odd by hypothesis, and we require $q \equiv 2 \pmod{r}$. The Chinese remainder theorem gives a reduced residue class satisfying these conditions. If there are no conditions, use the class $1 \pmod 2$. Dirichlet's theorem [Apo76] supplies infinitely many primes in this class. Choose one with $q > d + 1$, and let $m = (q-1)/d$.

Then $m > 1$, $m \mid q - 1$, and $m < q$. If $r \in \Pi \cap \operatorname{supp}(d)$, the chosen congruence gives $v_r(q-1) = a_r$, so $r \nmid m$. If $r \in \Pi \setminus \operatorname{supp}(d)$, then $q - 1 \equiv 1 \pmod{r}$, so again $r \nmid m$. The congruences also ensure that $r \nmid q$. Hence qm is divisible by no prime in Π.

Choose an integer t whose residue class has order m in $(\mathbb{Z}/q\mathbb{Z})^\times$, and let

$$A = \langle a, b \mid a^q = b^m = 1,\ bab^{-1} = a^t \rangle.$$

Since $m < q$, the subgroup $\langle a \rangle$ is the unique Sylow q-subgroup, hence characteristic. Every automorphism sends a to a^u with $u \in \mathbb{F}_q^\times$, and sends b to $a^v b^j$ with $v \in \mathbb{F}_q$ and $j \in (\mathbb{Z}/m\mathbb{Z})^\times$. Preserving the relation gives $t^j = t$, hence $j = 1$. Conversely, every choice of $u \in \mathbb{F}_q^\times$ and $v \in \mathbb{F}_q$ defines an automorphism by $a \mapsto a^u$ and $b \mapsto a^v b$. Indeed, $(a^v b)^m = 1$ because $1 + t + \cdots + t^{m-1} = 0$, and the conjugation relation is preserved. The images generate A. Thus $|\mathrm{Aut}(A)| = q(q-1)$, while $|A| = qm$, and $\alpha(A) = d$. □

For powers of $1/2$, we use the 2-primary part of the Schur multiplier of $\mathrm{PSL}_3(4)$. The required action of the outer automorphism group is recorded next.

Fact 2.13. *Let $L = \mathrm{PSL}_3(4)$. Then $M(L) \cong C_3 \times C_4 \times C_4$ and $\mathrm{Out}(L) \cong D_{12}$, with $|D_{12}| = 12$. Let $A \cong C_4^2$ be the 2-primary part of $M(L)$. The action of $\mathrm{Out}(L)$ on $A/2A$ has image $\mathrm{GL}_2(2)$. Its kernel is generated by the central involution of D_{12}, and this involution acts on A by inversion [BMO17].*

Theorem 2.14. *For every $e \geq 0$, there is a finite perfect group $D_{2,e}$ such that $\alpha(D_{2,e}) = 2^{-e}$ and $D_{2,e}/Z(D_{2,e})$ is a direct product of copies of $\mathrm{PSL}_3(4)$, with the empty product allowed. For $e \geq 3$, one can choose $D_{2,e}$ so that*

$$D_{2,e}/Z(D_{2,e}) \cong \mathrm{PSL}_3(4)^{3e+4},$$
$$Z(D_{2,e}) \cong C_2^{4e+4},$$
$$\mathrm{Out}(D_{2,e}) \cong C_2^{3e+4},$$
$$|D_{2,e}| = 20160^{3e+4} 2^{4e+4}.$$

Proof. For $e = 0$, take the trivial group. For $e = 2$, take the universal central extension $\widehat{L} \to L$. By Theorem 2.4 and Theorem 2.13, one has $\alpha(\widehat{L}) = 12/48 = 1/4$.

For $e = 1$, one has $M(L) = C_3 \times A$. Let $N = C_3 \times \langle 2a \rangle$ for some $a \in A$ of order 4. Then $|M(L)/N| = 8$. The C_3-factor is characteristic. The line stabilizer in $\mathrm{GL}_2(2)$ has order 2, so Theorem 2.13 gives a stabilizer of order 4 for $\langle a + 2A \rangle$ in $\mathrm{Out}(L)$. The map $A/2A \to 2A$, $x + 2A \mapsto 2x$, is an $\mathrm{Out}(L)$-equivariant

isomorphism. Consequently the stabilizer of $\langle 2a \rangle$ coincides with the stabilizer of $\langle a + 2A \rangle$, and has order 4. Hence Theorem 2.3 gives $\alpha(\widehat{L}/N) = 4/8 = 1/2$.

Now let $e \geq 3$, and let Γ_e be the graph in Theorem 2.10. Set $k = |E(\Gamma_e)| = 3e + 4$ and $v = |V(\Gamma_e)| = 2e + 4$. Let $Q = L^k$, and let $\widehat{Q} \to Q$ be its universal central extension. By Theorem 2.5, its multiplier is $(C_3 \times A)^k$. Choose a bijection between the k factors and $E(\Gamma_e)$, and identify $(A/2A)^k$ with $(\mathbb{F}_2^2)^E$. Let $U_{\Gamma_e} \leq (\mathbb{F}_2^2)^E$ be the graph code. Define $N_e = C_3^k \times \pi^{-1}(U_{\Gamma_e}) \leq (C_3 \times A)^k$, where π is reduction modulo 2 from A^k to $(A/2A)^k$.

By Theorem 2.11, the graph Γ_e has no isolated vertices, so F_{Γ_e} is injective. We obtain $|U_{\Gamma_e}| = 2^v$ and $|M(Q)/N_e| = 2^{2k-v}$. Simplicity of L gives $\mathrm{Out}(Q) \cong D_{12}^k \rtimes S_k$. The action on $(A/2A)^k$ has image $\mathrm{GL}_2(2)^k \rtimes S_k$ and kernel C_2^k. The inclusion of the C_3^k-part in N_e, together with the full inverse image $\pi^{-1}(U_{\Gamma_e})$, identifies the stabilizer of N_e with the preimage of the graph-code stabilizer. That target stabilizer is trivial by Theorems 2.9 and 2.11, leaving $\mathrm{Stab}_{\mathrm{Out}(Q)}(N_e) \cong C_2^k$.

Set $D_{2,e} = \widehat{Q}/N_e$. By Theorem 2.3, one has $Z(D_{2,e}) = M(Q)/N_e$ and $\mathrm{Out}(D_{2,e}) \cong C_2^k$. Thus $\alpha(D_{2,e}) = 2^k/2^{2k-v} = 2^{v-k} = 2^{-e}$. The center satisfies $Z(D_{2,e}) \cong (A/2A)^k/U_{\Gamma_e} \cong C_2^{2k-v} = C_2^{4e+4}$, and $D_{2,e}/Z(D_{2,e}) \cong Q$. The group $D_{2,e}$ is perfect because $\widehat{Q}$ is perfect. The cases $e = 1$ and $e = 2$ are also central quotients of perfect groups, and the trivial group handles $e = 0$. The usual order formula gives $|L| = 20160$ [Wil09], and the displayed order follows. □

For odd primes, we use Jacobi groups whose centers have order p and whose outer automorphism groups have order $p - 1$.

Let p be odd and let (V, ω) be a $2n$-dimensional symplectic vector space over $\mathbb{F}_p$, where $n \geq 1$. We use the symplectic-group facts in [Gro02]. Let $H(V) = V \times \mathbb{F}_p$, with multiplication $(v, a)(w, b) = (v + w, a + b + \omega(v, w)/2)$.

Lemma 2.15. *The group $H(V)$ has exponent p, and $Z(H(V)) = H(V)' = \{0\} \times \mathbb{F}_p$. If an automorphism of $H(V)$ acts trivially on both $H(V)/Z(H(V))$ and $Z(H(V))$, then it is inner.*

Proof. The commutator is $[(v, a), (w, b)] = (0, \omega(v, w))$. Nondegeneracy gives $Z(H(V)) = H(V)' = \{0\} \times \mathbb{F}_p$. The exponent statement follows from $(v, a)^m = (mv, ma)$. If θ acts trivially on the quotient and the center, then $\theta(v, a) = (v, a + \lambda(v))$ for a function λ from V to $\mathbb{F}_p$. The homomorphism property makes λ additive, and hence $\mathbb{F}_p$-linear. Nondegeneracy gives $\lambda(v) = \omega(u, v)$ for some $u \in V$. Conjugation by $(u, 0)$ sends (v, a) to $(v, a + \omega(u, v))$, so θ is inner. □

Let $S = \mathrm{Sp}(V, \omega)$, acting on $H(V)$ by $s(v, a) = (sv, a)$, and let $J_{p,n} = H(V) \rtimes S$. Since this action is trivial on the second coordinate of $H(V)$, the subgroup $Z(H(V)) = \{0\} \times \mathbb{F}_p$ remains central in $J_{p,n}$.

Proposition 2.16. *For every odd prime p and every $n \geq 3$, the group $J_{p,n}$ is perfect, $Z(J_{p,n}) \cong C_p$, and $\mathrm{Out}(J_{p,n}) \cong \mathbb{F}_p^\times$. Hence $\alpha(J_{p,n}) = (p - 1)/p$. The quotient $J_{p,n}/Z(J_{p,n})$ is centerless and has a unique minimal normal subgroup, namely $H(V)/Z(H(V)) \cong V$.*

Proof. Denote $H = H(V)$, $J = J_{p,n}$, and $Z = Z(H)$. For $n \geq 3$, the group S is perfect, $O_p(S) = 1$, and acts irreducibly and faithfully on V. Since $[H, S]$ maps onto V modulo Z, and $H' = Z$, the group J is perfect. The quotient $J/Z \cong V \rtimes S$ is centerless by faithfulness and $V^S = 0$, so $Z(J) = Z$.

The subgroup H is $O_p(J)$, hence characteristic. Let $\varphi \in \mathrm{Aut}(J)$. It induces $T \in \mathrm{GL}(V)$, a scalar $c \in \mathbb{F}_p^\times$ on Z, and an automorphism β of $J/H \cong S$. The commutator pairing gives $\omega(Tv, Tw) = c\omega(v, w)$, so $T \in \mathrm{GSp}(V, \omega)$. For $s \in S$, compatibility of conjugation on H/Z gives $Ts = \beta(s)T$, with S viewed in its natural representation. Thus $\beta(s) = TsT^{-1}$.

Conversely, every $T \in \mathrm{GSp}(V, \omega)$ defines an automorphism by $(v, a) \mapsto (Tv, \mu(T)a)$ and $s \mapsto TsT^{-1}$, where $\mu(T)$ is the similitude factor. Compose φ with the inverse of this automorphism. The result is trivial on H/Z, Z, and J/H. By Theorem 2.15, its restriction to H is conjugation by some $h \in H$. Removing this inner automorphism leaves an automorphism that fixes H pointwise and sends s to $\delta(s)s$ with $\delta(s) \in H$. Each $\delta(s)$ centralizes H and therefore lies in Z. The resulting homomorphism δ from S to Z is trivial because S is perfect.

Thus every automorphism of J is a product of conjugation by an element of H and a similitude automorphism. If $T \in \mathrm{Sp}(V, \omega)$, then the similitude automorphism defined by T is conjugation by $(0, T) \in J$, so it is inner. Conversely, if the similitude automorphism defined by T is conjugation by some $(h, s_0) \in J$, then the two automorphisms induce the same map on H/Z, and hence $T = s_0 \in \mathrm{Sp}(V, \omega)$. The kernel in the outer quotient is therefore exactly $\mathrm{Sp}(V, \omega)$. Hence $\mathrm{Out}(J) \cong \mathrm{GSp}(V, \omega)/\mathrm{Sp}(V, \omega) \cong \mathbb{F}_p^\times$. The formula for $\alpha(J)$ follows from Theorem 2.1.

In $V \rtimes S$, the subgroup V is minimal normal by irreducibility. Let $N \trianglelefteq V \rtimes S$ be nontrivial. If $N \leq V$, then $N = V$. Otherwise choose $(v, s) \in N$ with $s \neq 1$. Since S acts faithfully on V, some $w \in V$ satisfies $sw \neq w$, and the commutator of (v, s) with $(w, 1)$ is the nontrivial element $(sw - w, 1)$ of $N \cap V$. Hence $V \leq N$. Therefore V is the unique minimal normal subgroup. □

2.4 Realization by finite groups

A factor of ratio $1/p$ would be convenient, but it is not needed. The Jacobi factor in Theorem 2.16 has ratio $(p-1)/p$, and every prime divisor of $p - 1$ is smaller than p. Its numerator can therefore be removed recursively using factors associated with smaller primes. For example,

$$\frac{1}{16}\left(\frac{2}{3}\right)^2 \frac{36}{37} = \frac{1}{37}.$$

Accordingly, $D_{2,4} \times J_{37,n} \times J_{3,n_1} \times J_{3,n_2}$ has ratio $1/37$, provided $n, n_1, n_2 \geq 3$ and $n_1 \neq n_2$. The inequality prevents the two 3-factors from being permuted by automorphisms.

Proposition 2.17. *For every positive integer b, there is a finite perfect group D_b with $\alpha(D_b) = 1/b$.*

Proof. Express b as $b = \prod_p p^{e_p}$, with $e_p = 0$ outside $\mathrm{supp}(b)$. Let P be the largest prime divisor of b, with $P = 2$ if $b = 1$. For each prime $p \leq P$, define x_p, in descending order, by

$$x_p = e_p + \sum_{\substack{\ell > p \\ \ell \leq P \\ \ell \text{ prime}}} x_\ell v_p(\ell - 1),$$

where an empty sum is zero.

For every odd $p \leq P$, choose pairwise distinct integers $n(p, i) \geq 3$, with $1 \leq i \leq x_p$, so that the orders

$p^{2n(p,i)}$ are distinct among all chosen Jacobi factors. Let

$$D_b = D_{2,x_2} \times \prod_{\substack{p \le P \\ p \text{ odd}}} \prod_{i=1}^{x_p} J_{p,n(p,i)}.$$

Let $\overline{D}_b = D_b/Z(D_b)$, let $\overline{B} = D_{2,x_2}/Z(D_{2,x_2})$, and let I be the set of chosen pairs (p,i). We show that every automorphism of $\overline{D}_b$ preserves its displayed direct factors. For $(p,i) \in I$, let $\overline{J}_{p,i} = J_{p,n(p,i)}/Z(J_{p,n(p,i)})$, and let $V_{p,i}$ be its unique minimal normal subgroup from Theorem 2.16. It is abelian of order $p^{2n(p,i)}$, and these orders are distinct by construction.

Let R be a minimal normal subgroup of $\overline{D}_b$. For each displayed direct factor X, minimality gives either $[R,X] = 1$ or $[R,X] = R$. The first alternative cannot hold for every X, since $\overline{D}_b$ is centerless. Thus $R = [R,X] \le X$ for some factor X. It follows from Theorems 2.14 and 2.16 that the minimal normal subgroups of $\overline{D}_b$ are precisely the simple factors of $\overline{B}$ and the groups $V_{p,i}$. The product of the nonabelian simple minimal normal subgroups is $\overline{B}$, while the abelian minimal normal subgroups $V_{p,i}$ have pairwise distinct orders. Hence $\overline{B}$ and every $V_{p,i}$ are characteristic.

For each $(p,i) \in I$, the factor $\overline{J}_{p,i}$ is recovered as

$$\overline{J}_{p,i} = C_{\overline{D}_b}\left(\overline{B} \prod_{\substack{(q,j)\in I \\ (q,j)\neq(p,i)}} V_{q,j}\right)'. \tag{2.1}$$

Here $C_{\overline{B}}(\overline{B}) = 1$, while $C_{\overline{J}_{q,j}}(V_{q,j}) = V_{q,j}$ by faithfulness. The centralizer in Equation (2.1) is therefore $\overline{J}_{p,i} \times \prod_{(q,j)\neq(p,i)} V_{q,j}$, whose derived subgroup is $\overline{J}_{p,i}$. Every automorphism of $\overline{D}_b$ consequently preserves $\overline{B}$ and all $\overline{J}_{p,i}$, so Theorem 2.6 applies.

Consequently $\alpha(D_b) = 2^{-x_2} \prod_{p\le P,\ p \text{ odd}} ((p-1)/p)^{x_p}$. For an odd prime $q \le P$, one has

$$v_q(\alpha(D_b)) = -x_q + \sum_{\substack{\ell > q \\ \ell \le P \\ \ell \text{ prime}}} x_\ell v_q(\ell - 1) = -e_q.$$

The same calculation at 2 gives $v_2(\alpha(D_b)) = -e_2$. If $q > P$, then q divides neither a denominator $p \le P$ nor any $\ell - 1$ with $\ell \le P$. Hence $v_q(\alpha(D_b)) = 0$. Therefore $\alpha(D_b) = 1/b$. The group D_b is perfect because all its factors are perfect. □

Theorem 2.18. *For every $r \in \mathbb{Q}^+$, there is a finite group G such that $\alpha(G) = r$. More precisely, if $r = a/b$ with $a, b \in \mathbb{Z}^+$, then G can be chosen as a coprime direct product $A \times B$, with A metacyclic and chosen to avoid the primes dividing $|B|$, and with B perfect.*

Proof. Choose a perfect group B with $\alpha(B) = 1/(2b)$ by Theorem 2.17, and let Π be the set of primes dividing $|B|$. Since $|B| = 2b|\mathrm{Aut}(B)|$, the prime 2 belongs to Π. Taking $d = 2a$ satisfies the parity hypothesis in Theorem 2.12. We obtain A with $\alpha(A) = 2a$ and $\gcd(|A|, |B|) = 1$. By Theorem 2.7, one has $\alpha(A \times B) = \alpha(A)\alpha(B) = a/b$. □

3 A nilpotent construction

Let $N = \prod_p P_p$ be finite and nilpotent. Its Sylow subgroups are characteristic, so $\alpha(N) = \prod_p \alpha(P_p)$. The difficulty is not multiplicativity but compatibility among the local choices: a value $m_p p^k$ changes the valuations at every prime dividing m_p, not only the valuation at p. We must therefore ensure that closing the prescribed support under $p \mapsto \operatorname{supp}(m_p)$ introduces only finitely many primes.

3.1 A local-to-global criterion

Theorem 3.1. *Suppose that for every prime p there is a positive integer m_p, coprime to p, such that $m_p p^{\mathbb{Z}} \subseteq \mathcal{I}_p$. Suppose also that there is a finite set of primes $\mathcal{E}$, independent of p, such that $\operatorname{supp}(m_p) \subseteq \mathcal{E} \cup \{q \mid q < p\}$. Then finite nilpotent groups realize every positive rational number.*

Proof. Write $r = \prod_q q^{a_q}$, with only finitely many nonzero exponents, and let S be the smallest set of primes containing $\operatorname{supp}(r)$ and satisfying

$$p \in S \quad \Longrightarrow \quad \operatorname{supp}(m_p) \subseteq S.$$

If $S = \varnothing$, then $r = 1$, which is realized by the trivial group. Otherwise, let $B = \max(\mathcal{E} \cup \operatorname{supp}(r))$. Every prime introduced while forming S is at most B: a new prime either belongs to $\mathcal{E}$, or is smaller than a prime already present. Hence S is finite.

For each $q \in S$, set

$$k_q = a_q - \sum_{p \in S} v_q(m_p).$$

Choose a finite q-group P_q with $\alpha(P_q) = m_q q^{k_q}$, and let $N = \prod_{q \in S} P_q$. By Theorem 2.7,

$$\alpha(N) = \prod_{q \in S} \alpha(P_q).$$

For $q \in S$, the q-adic valuation of this product is

$$k_q + \sum_{p \in S} v_q(m_p) = a_q.$$

If $q \notin S$, then $a_q = 0$, and the closure property gives $v_q(m_p) = 0$ for every $p \in S$. Thus $\alpha(N) = r$. □

3.2 The González-Sánchez-Jaikin-Zapirain tower

The local construction starts with uniform pro-p groups whose finite quotients have bounded outer automorphism groups. Their abelianizations eventually stabilize, while their centers acquire a cyclic factor of order p^i. Consequently their automorphism ratios have a fixed prime-to-p part and p-adic exponents tending to $-\infty$.

Fact 3.2. *For every prime p, there is a uniform pro-p group $U = U_p$ [GJ15]. Set $U_i = U/U^{p^i}$. Then $\operatorname{Aut}(U) \cong \varprojlim_i \operatorname{Aut}(U_i)$, and the groups $\operatorname{Out}(U_i)$ have bounded order [GJ15]. Let $L = \log(U)$ [Dix+99]. Then $[L, L] \subseteq p^2 L$ and $L \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong \mathfrak{M} \otimes_{\mathbb{Q}} \mathbb{Q}_p$, where $\mathfrak{M}$ is Sato's algebra. The algebra $\mathfrak{M}$ has dimension 41, has one-dimensional center and only inner derivations, is perfect, and has characteristic*

nilradical $\mathfrak{n}$ *with* $\mathfrak{M}/\mathfrak{n} \cong \mathfrak{sl}_2$ *[Sat71].*

Lemma 3.3. *For every prime* p*, there are finite abelian* p*-groups* A_p *and* T_p *such that, for all sufficiently large* i*, one has* $U_i^{\mathrm{ab}} \cong A_p$ *and* $Z(U_i) \cong C_{p^i} \times T_p$*.*

Proof. Consider first the abelianization. Let $L = \log(U)$, let $\mathfrak{g} = L \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$, and let $D = \overline{[U,U]}$. Denote the Lie algebra of D by $\mathfrak{d}$. The quotient U/D is abelian, so $\mathfrak{g}/\mathfrak{d}$ is abelian. Perfectness of $\mathfrak{g}$ forces $\mathfrak{d} = \mathfrak{g}$, which makes D open in U [Dix+99]. Let $A_p = U/D$. For sufficiently large i, one has $U^{p^i} \leq D$. The subgroup $[U,U]U^{p^i}$ is then open and dense in D, and hence equals D. Therefore $U_i^{\mathrm{ab}} \cong U/([U,U]U^{p^i}) = A_p$.

For the center, let $Z_L = L \cap Z(\mathfrak{g})$. This is a saturated rank-one $\mathbb{Z}_p$-submodule of L. Consider the map Θ from L/Z_L to $\mathrm{End}_{\mathbb{Z}_p}(L)$ given by $x + Z_L \mapsto \mathrm{ad}\, x$. The map is injective. For $i \geq 1$, let $C_i = \{x \in L \mid [x, L] \subseteq p^i L\}$. Since $U^{p^i} = \exp(p^i L)$, the map $x + p^i L \mapsto \exp(x)U^{p^i}$ gives a bijection $C_i/p^i L \to Z(U_i)$. Indeed, $\exp(x)$ is central modulo U^{p^i} exactly when $(e^{\mathrm{ad}\, x} - 1)L \subseteq p^i L$. Since $\mathrm{ad}\, x \in p^2 \mathrm{End}_{\mathbb{Z}_p}(L)$, one has $e^T - 1 = Th(T)$, with $T = \mathrm{ad}\, x$. Here $h(T) \equiv I \pmod{p}$, so $h(T) \in \mathrm{End}_{\mathbb{Z}_p}(L)^\times$. Centrality is therefore equivalent to $[x, L] \subseteq p^i L$.

Since L/Z_L has rank 40, Smith normal form applied to Θ gives integers $s_1, \ldots, s_{40} \geq 0$ such that, for $i \geq \max_j s_j$, the additive group $C_i/p^i L$ is $C_{p^i} \times \prod_{j=1}^{40} C_{p^{s_j}}$, with trivial factors omitted. For odd p, the BCH correction terms for elements of C_i lie in $p^i L$, and the preceding bijection is a group isomorphism. For $p = 2$, the bijection need not respect the BCH product. We instead compare torsion cardinalities. For every $r \geq 0$, the identity $(\exp x)^{p^r} = \exp(p^r x)$ gives $|Z(U_i)[p^r]| = |(C_i/p^i L)[p^r]|$. These cardinalities determine a finite abelian p-group, so $Z(U_i) \cong C_{p^i} \times T_p$, where $T_p = \prod_j C_{p^{s_j}}$, with trivial factors omitted. □

Denote $|A_p|$ by p^{a_p} and $|T_p|$ by p^{t_p}.

Lemma 3.4. *For every prime* p*, the prime-to-*p *part* $m_i = |\mathrm{Out}(U_i)|_{p'}$ *is eventually constant. Denote the eventual value by* m_p*.*

Proof. For $i \geq j \geq 1$, the characteristic subgroup U^{p^j}/U^{p^i} gives a homomorphism from $\mathrm{Aut}(U_i)$ to $\mathrm{Aut}(U_j)$. Its kernel acts trivially on $U_i/\Phi(U_i)$, because $U^{p^j}/U^{p^i} \leq \Phi(U_i)$. The Frattini-quotient theorem makes this kernel a p-group [Gor80]. It follows that $|\mathrm{Aut}(U_i)|_{p'}$ divides $|\mathrm{Aut}(U_j)|_{p'}$. Inner automorphisms contribute no prime divisor other than p, so $|\mathrm{Out}(U_i)|_{p'} = |\mathrm{Aut}(U_i)|_{p'}$. These positive integers form a descending chain under divisibility and must stabilize. □

3.3 Vertical lines in $\mathcal{I}_p$

The groups U_i already give values $m_p p^{k_i}$ with k_i unbounded below. They do not, however, necessarily give every sufficiently negative exponent, since the integers k_i need not be consecutive.

Lemma 3.5. *For every odd prime* p *and every* $n \geq 3$*, let*

$$H_{p,n} = \langle a, b \mid a^{p^n} = 1,\ b^{p^n} = a^{p^{n-1}},\ b^{-1}ab = a^{1+p} \rangle.$$

Then $H_{p,n}$ is indecomposable, $\alpha(H_{p,n}) = p$, $H_{p,n}^{\mathrm{ab}} \cong C_p \times C_{p^n}$, and $Z(H_{p,n}) \cong C_{p^2}$.

Proof. This is Menegazzo's metacyclic case $m = l = n$, $s = 1$, and $h = n - 1$, with generators interchanged [Men93]. In particular $H_{p,n}$ has order p^{2n}, the normal form $a^i b^j$, and $|\mathrm{Aut}(H_{p,n})| = p^{2n+1}$. Hence $\alpha(H_{p,n}) = p$.

In the abelianization, $b^{-1}ab = a^{1+p}$ gives $\bar{a}^p = 1$, while $b^{p^n} = a^{p^{n-1}}$ gives $\bar{b}^{p^n} = 1$. The quotient $a \mapsto (1,0)$, $b \mapsto (0,1)$ onto $C_p \times C_{p^n}$ gives $H_{p,n}^{\mathrm{ab}} \cong C_p \times C_{p^n}$.

Using the normal form, $a^i b^j$ centralizes a exactly when $p^{n-1} \mid j$, and centralizes b exactly when $p^{n-1} \mid i$. Thus $Z(H_{p,n}) = \langle b^{p^{n-1}} \rangle \cong C_{p^2}$, since $(b^{p^{n-1}})^p = a^{p^{n-1}} \neq 1$. A finite p-group with cyclic center is indecomposable. □

If X and Y are finite groups with no common nontrivial direct factor, the formula of Bidwell, Curran, and McCaughan [BCM06] gives

$$\alpha(X \times Y) = \alpha(X)\alpha(Y)|\mathrm{Hom}(X, Z(Y))|\,|\mathrm{Hom}(Y, Z(X))|. \tag{3.1}$$

We also use the uniqueness of indecomposable direct factors in the Krull-Remak-Schmidt theorem [Rob96].

Lemma 3.6. *For every prime p, there is an integer c_p such that $m_p p^k \in \mathcal{I}_p$ for all $k \leq c_p$.*

Proof. On the common tail supplied by Theorems 3.3 and 3.4, let $|\mathrm{Out}(U_i)| = m_p p^{b_i}$. The integers b_i are bounded because the outer automorphism groups in the González-Sánchez and Jaikin-Zapirain tower have bounded order [GJ15]. After passing to an infinite set of indices, we may assume that $b_i = b_p$ is constant. Then $\alpha(U_i) = m_p p^{b_p - i - t_p}$.

First assume that p is odd. Choose $E_p \geq 3$ so that $p^{E_p} > \exp(A_p)$ and $p^{E_p} \geq \exp(T_p)$. Let $E_p \leq n \leq i$. Since $H_{p,n}$ is indecomposable, any common nontrivial direct factor of U_i and $H_{p,n}$ would be $H_{p,n}$. Then $H_{p,n}^{\mathrm{ab}}$ would be a direct factor of A_p, impossible by the choice of n. Thus Equation (3.1) applies.

Let $u_p = \log_p|\mathrm{Hom}(A_p, C_{p^2})|$, and let $\rho_p = \log_p|T_p[p]|$. Then $|\mathrm{Hom}(U_i, Z(H_{p,n}))| = p^{u_p}$, while $Z(U_i) \cong C_{p^i} \times T_p$ and $H_{p,n}^{\mathrm{ab}} \cong C_p \times C_{p^n}$ give $|\mathrm{Hom}(H_{p,n}, Z(U_i))| = p^{n+1+t_p+\rho_p}$. Hence $\alpha(U_i \times H_{p,n}) = m_p p^{n-i+c_p}$, where $c_p = b_p + u_p + \rho_p + 2$. Given $k \leq c_p$, choose i in the chosen infinite set so large that $n = i + k - c_p \geq E_p$. This realizes $m_p p^k$.

For $p = 2$, choose $E_2 \geq 1$ so that $2^{E_2} > \exp(A_2)$ and $2^{E_2} \geq \exp(T_2)$. Let $E_2 \leq s \leq i$. The groups U_i and C_{2^s} have no common nontrivial direct factor by the same abelianization argument. Since $\alpha(C_{2^s}) = 1/2$, $|\mathrm{Hom}(U_i, C_{2^s})| = 2^{a_2}$, and $|\mathrm{Hom}(C_{2^s}, Z(U_i))| = 2^{s+t_2}$, one obtains $\alpha(U_i \times C_{2^s}) = m_2 2^{s-i+c_2}$, where $c_2 = b_2 + a_2 - 1$. Choosing $s = i + k - c_2$, with i sufficiently large in the chosen infinite set, realizes every $k \leq c_2$. □

Lemma 3.7. *For every prime p, there are finite p-groups Q with $\alpha(Q)$ a positive power of p of arbitrarily large exponent. They may be chosen so that their indecomposable direct factors have abelianization exponent larger than any prescribed bound.*

Proof. For odd p, choose $n_1 < \cdots < n_r$, all above the prescribed bound, and let $Q = \prod_{j=1}^{r} H_{p,n_j}$. The strict increase of the abelianization exponents rules out a common nontrivial direct factor at each stage by the Krull-Remak-Schmidt theorem [Rob96]. Repeated application of Equation (3.1) gives $\alpha(Q) = p^{\tau(Q)}$. Each factor contributes one power of p, and every additional Hom term is a p-power. Hence $\tau(Q) \geq r$, which makes the exponent unbounded as r increases.

For $p = 2$, let $Q = C_{2^{\lambda_1}} \times \cdots \times C_{2^{\lambda_r}}$, with $1 \leq \lambda_1 < \cdots < \lambda_r$. The automorphism order formula for finite abelian groups [HR07] gives

$$\alpha(Q) = 2^{2\sum_{a<b} \min(\lambda_a, \lambda_b) - r}.$$

For $r = 2$, this exponent is $2\lambda_1 - 2$, so positive exponents of arbitrarily large size already occur. In the odd case, all of the n_j, and in the case $p = 2$, all of the λ_i, may be taken larger than any prescribed bound. □

Theorem 3.8. *For every prime p, the stable integer m_p from Theorem 3.4 satisfies $m_p p^{\mathbb{Z}} \subseteq \mathcal{I}_p$.*

Proof. Start in the range already supplied by Theorem 3.6. For $j \leq c_p$ and every sufficiently large selected index i, its proof gives a group $R_{i,j} = U_i \times W_d$ with $\alpha(R_{i,j}) = m_p p^j$, where $d = i + j - c_p$, with $W_d = H_{p,d}$ for odd p and $W_d = C_{2^d}$ for $p = 2$.

Select Q as in Theorem 3.7, with every indecomposable direct factor having abelianization exponent larger than $\exp(A_p)$. This excludes such a factor from U_i, whose abelianization is A_p. Choose i large enough that p^d also exceeds the abelianization exponent of every indecomposable direct factor of Q. None of these factors is then isomorphic to W_d. The Krull-Remak-Schmidt theorem now shows that $R_{i,j}$ and Q have no common nontrivial direct factor [Rob96], so Equation (3.1) applies.

Choose $E(Q)$ so that $p^{E(Q)}$ is at least $\exp(T_p)$, $\exp Z(Q)$, and $\exp(Q^{\mathrm{ab}})$, and so that $E(Q)$ is at least the lower bound used in Theorem 3.6. If $i, d \geq E(Q)$, then $U_i^{\mathrm{ab}} \cong A_p$, $Z(U_i) \cong C_{p^i} \times T_p$, and the exponent bounds make the two Hom factors in Equation (3.1) independent of i and j. Hence there is an integer $\lambda(Q)$ such that $\alpha(R_{i,j} \times Q) = m_p p^{j+\lambda(Q)}$. Since the Hom factors are p-powers, $\lambda(Q) \geq v_p(\alpha(Q))$. By Theorem 3.7, this lower bound is arbitrarily large. Given $k \in \mathbb{Z}$, choose Q with $c_p + \lambda(Q) \geq k$, set $j = k - \lambda(Q)$, and choose i large enough for the preceding conditions. Then $\alpha(R_{i,j} \times Q) = m_p p^k$. □

3.4 Prime divisors of m_p

It remains to control the primes $q \neq p$ dividing m_p. Such a prime gives an automorphism of order q in the inverse limit, and hence an automorphism of the associated rational Lie algebra. Its action on the $\mathfrak{sl}_2$-quotient will force $q < p$, apart from a fixed finite set of exceptions.

Lemma 3.9. *Let $q \neq p$ be a prime dividing m_p. Then $\mathrm{Aut}(U_p)$ contains an element of order q, and hence $\mathrm{Aut}_{\mathbb{Q}_p}(\mathfrak{M} \otimes_{\mathbb{Q}} \mathbb{Q}_p)$ contains an element of order q.*

Proof. Work on a cofinal tail of indices for which q divides $|\mathrm{Out}(U_i)|$, and hence $|\mathrm{Aut}(U_i)|$. By Cauchy's theorem, the set Ω_i of elements of order q in $\mathrm{Aut}(U_i)$ is nonempty. For $i \geq j$ in this tail, the kernel of $\mathrm{Aut}(U_i) \to \mathrm{Aut}(U_j)$ is a p-group by the Frattini-quotient theorem used in Theorem 3.4 [Gor80], so reduction sends Ω_i into Ω_j.

The maps $\Omega_i \to \Omega_j$ need not be surjective. Nevertheless, every finite collection of compatibility conditions can be satisfied: choose one common higher level, take an element of order q there, and reduce it to the selected levels. The reductions still have order q, because the kernels are p-groups and $q \neq p$. Thus the closed compatibility conditions in the compact product $\prod_i \Omega_i$ have the finite-intersection property, and their intersection contains a compatible system. Since the tail is cofinal and $\mathrm{Aut}(U_p) \cong \varprojlim_i \mathrm{Aut}(U_i)$ by Theorem 3.2, this system defines an element of order q in $\mathrm{Aut}(U_p)$. Its image under the logarithm functor is an automorphism of the rational Lie algebra of the same order. □

Let $A = \mathrm{Aut}_{\mathbb{Q}}(\mathfrak{M})$. Since the nilradical $\mathfrak{n}$ is characteristic, there is a homomorphism of algebraic groups π from A to $\mathrm{Aut}(\mathfrak{M}/\mathfrak{n}) \cong \mathrm{PGL}_2$. Let $K = \ker \pi$.

Lemma 3.10. *The identity component K° is unipotent.*

Proof. Since $\mathrm{Der}(\mathfrak{M}) = \mathrm{ad}(\mathfrak{M})$, and since $\mathfrak{M}/\mathfrak{n} \cong \mathfrak{sl}_2$ is centerless, one has $\mathrm{Lie}(K) = \mathrm{ad}(\mathfrak{n})$. For $x \in \mathfrak{n}$, the endomorphism $\mathrm{ad}\, x$ sends $\mathfrak{M}$ into $\mathfrak{n}$ and is nilpotent on $\mathfrak{n}$ by Engel's theorem [Hum72]. Hence it is nilpotent on $\mathfrak{M}$, so every element of $\mathrm{Lie}(K)$ is nilpotent in the ambient matrix representation.

After extension to an algebraic closure, Engel's theorem realizes $\mathrm{Lie}(K^\circ)$ by strictly upper triangular matrices [Hum72]. The group K° is therefore solvable and is triangularizable by Lie-Kolchin [Hum75]. The diagonal characters have zero differential because the Lie algebra acts nilpotently. A character with zero differential is trivial on the connected group K°, so K° is unipotent. □

Let $e_0 = |K_{\overline{\mathbb{Q}}}/K^\circ_{\overline{\mathbb{Q}}}|$, and let $\mathcal{E} = \{2, 3\} \cup \mathrm{supp}(e_0)$.

Theorem 3.11. *For every prime p, one has $\mathrm{supp}(m_p) \subseteq \mathcal{E} \cup \{q \mid q < p\}$.*

Proof. Let $q \mid m_p$ with $q \notin \mathcal{E}$. Since m_p is coprime to p, one has $q \neq p$. By Theorem 3.9, there is an element $\sigma \in A(\mathbb{Q}_p)$ of order q. If $\pi(\sigma) = 1$, then $\sigma \in K(\mathbb{Q}_p)$. After base change to $\overline{\mathbb{Q}}_p$, the geometric component group again has order e_0. Hence the image of σ in it has order dividing both q and e_0. Since $q \nmid e_0$, this image is trivial, so $\sigma \in K^\circ(\overline{\mathbb{Q}}_p)$. This is impossible because a unipotent group over a characteristic zero field has no nontrivial finite-order elements. Thus $\pi(\sigma)$ has order q in $\mathrm{PGL}_2(\mathbb{Q}_p)$.

Choose a representative $T \in \mathrm{GL}_2(\mathbb{Q}_p)$, and let λ, μ be its eigenvalues over a splitting field. The relation $T^q = cI$ shows that $\zeta = \lambda/\mu$ is a qth root of unity. It is nontrivial, since equal eigenvalues would make the projective class either trivial or nontrivial unipotent, and a nontrivial unipotent class has infinite order in characteristic zero. As q is prime, ζ is primitive. The field $\mathbb{Q}_p(\zeta)$ lies in the splitting field of a quadratic polynomial, so its residue degree f is at most 2. This degree bound is the one place where the two-dimensional representation behind PGL_2 matters. Reduction modulo the maximal ideal is injective on qth roots of unity because $q \neq p$. We obtain $q \mid p^f - 1$, with $f \leq 2$, and therefore $q \mid p - 1$ or $q \mid p + 1$. If p is odd, every odd prime divisor of $p - 1$ or $p + 1$ is less than p, and the possible divisor 2 lies in $\mathcal{E}$. If $p = 2$, the only possible odd prime is 3, also in $\mathcal{E}$. This proves the claim. □

Together, Theorems 3.8 and 3.11 prove Theorem 1.2.

3.5 Realization by finite nilpotent groups

Theorem 3.12. *For every $r \in \mathbb{Q}^+$, there is a finite nilpotent group N with $\alpha(N) = r$. Equivalently,*

$$\{\alpha(N) \mid N \text{ finite nilpotent}\} = \mathbb{Q}^+.$$

Proof. The hypotheses of Theorem 3.1 are given by Theorem 3.8 and Theorem 3.11. Hence Theorem 3.1 gives $\mathbb{Q}^+ \subseteq \{\alpha(N) \mid N \text{ finite nilpotent}\}$. The reverse inclusion is automatic, since $\alpha(N)$ is a positive rational for every finite group N. □

Corollary 3.13. *The images of α on finite supersolvable groups and on finite solvable groups are both $\mathbb{Q}^+$.*

Proof. Every finite nilpotent group is supersolvable, and every finite supersolvable group is solvable. Hence Theorem 3.12 gives the result. □

4 Infinite fibers

Surjectivity does not determine the size of a fiber. To construct further groups without changing α, we adjoin complete simple direct factors. Their ratio is 1, and choosing new composition factors prevents automorphisms from mixing them with the group already constructed.

Lemma 4.1. *Let G be finite, and let S be finite, centerless, simple, and complete. If S is not a composition factor of G, then $\alpha(G \times S) = \alpha(G)$.*

Proof. Let $\varphi \in \mathrm{Aut}(G \times S)$, and let $N = \varphi(1 \times S)$. Then $N \trianglelefteq G \times S$ and $N \cong S$. If the projection of N to G were nontrivial, its kernel would be a proper normal subgroup of the simple group N, so the projection would be injective. Its image would then be a normal subgroup of G isomorphic to S, contrary to the composition-factor hypothesis. The projection is therefore trivial, and $N = 1 \times S$. Its centralizer is $G \times 1$ because $Z(S) = 1$. Both direct factors are characteristic, giving $\mathrm{Aut}(G \times S) \cong \mathrm{Aut}(G) \times \mathrm{Aut}(S)$. Completeness gives $\alpha(S) = 1$. □

Theorem 4.2. *For every $r \in \mathbb{Q}^+$, the set of isomorphism classes of finite groups G with $\alpha(G) = r$ is countably infinite. Representatives may be chosen with unbounded order and with arbitrarily many nonabelian composition factors.*

Proof. Choose G_0 with $\alpha(G_0) = r$ by Theorem 2.18. Having chosen G_{i-1}, choose a finite centerless complete simple group S_i with $|S_i| > |G_{i-1}|$. For example, the groups of adjoint type $E_8(p)$, with p prime, are centerless, simple, and complete [Wil09]. Set $G_i = G_{i-1} \times S_i$. The order inequality ensures that S_i is not a composition factor of G_{i-1}, so Theorem 4.1 gives $\alpha(G_i) = r$. The orders $|G_i|$ strictly increase, and each step adds a nonabelian composition factor. Thus the fiber over r contains groups of unbounded order and groups with arbitrarily many nonabelian composition factors. It is countable because there are only countably many finite groups up to isomorphism. □

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